1 **(a)** $-15 \leq 3x$ or $-3x \leq 15$ **[M1]**

$x \geq -5$ or $-5 \leq x$ **[A1]**

(b) $-7 < 2x \leq 8$ **[M1]**

$-3.5 < x \leq 4$ **[A1]**

(c) Largest $= 4$ **[B1]**

Smallest $= -3$ **[B1]**

[Total Marks: 6]

2 **(a)** $2x < 12$ **[M1]**

$x < 6$ **[A1]**

(b) Recognises open circle means strict inequality ($>$ or $<$) and arrow to right means greater than **[M1]**

$x > -2$ **[A1]**

[Total Marks: 4]

3 Identifies horizontal line $y = 2$ **[M1]**

States $y \geq 2$ (accept $y > 2$) **[A1]**

Identifies dashed line passing through $(0, 6)$ and $(4, 2)$ **[M1]**

Finds equation $y = -x + 6$ and states $y < -x + 6$ **[A1]**

Identifies solid line passing through $(0, 2)$ and $(1, 5)$ **[M1]**

Finds equation $y = 3x + 2$ and states $y \leq 3x + 2$ **[A1]**

[Total Marks: 6]

4 $y \geq 2$ **[B1]**

$y \leq 2x$ **[B1]**

$x + y \leq 6$ **[B1]**

[Total Marks: 3]

5 **(a)** $x + y \leq 500$ **[B1]**

$y \geq 150$ **[B1]**

$5x + 10y \geq 3000$ (or simplified to $x + 2y \geq 600$) **[B1]**

(b) Line drawn correctly for $x + y = 500$ (passing through $(0, 500)$ and $(500, 0)$) **[B1]**

Line drawn correctly for $y = 150$ (horizontal line) **[B1]**

Line drawn correctly for $x + 2y = 600$ (passing through $(0, 300)$ and $(600, 0)$) **[B1]**

Correct region identified and all unwanted regions properly shaded (leaving quadrilateral with vertices $(0, 300), (0, 500), (350, 150), (300, 150)$ unshaded) **[B1]**

(c) Identifies the vertex $(350, 150)$ as giving the maximum x value **[M1]**

Children's tickets = 350, Adult tickets = 150 **[A1]**

[Total Marks: 9]

6 **(a)** Adding 1 to all parts: $-6 \leq 3x < 12$ **[M2, M1 for one side correct]**

Dividing by 3: $-2 \leq x < 4$ **[A1]**

(b) Identifies at least 3 correct integers or includes 4 but excludes -2 **[M1]**

$-2, -1, 0, 1, 2, 3$ **[A1]**

[Total Marks: 5]